Crime Analytics Dashboard

1. Project Overview

This project analyses crime reports using a large public dataset containing over **750,000 crime records**. The goal is to identify patterns in crime occurrences, victim demographics, weapon usage, and geographic distribution to support data-driven insights.

The analysis combines **Python data preprocessing and Power BI visualization** to transform raw crime data into an interactive analytics dashboard.

2. Dataset Summary

Rows:

752,000 crime records

Columns:

20+ features

Key attributes include:

- Crime description
- Area name
- Victim age
- Victim gender
- Weapon used
- Premises description
- Crime occurrence date and time

3. Data Cleaning & Preparation (Python)

Data preprocessing was performed using **Python (Pandas)**.

Key steps included:

• Handling missing values in victim demographic fields
• Standardizing column names into snake case
• Cleaning weapon and premises descriptions
• Converting date columns into datetime format
• Creating derived columns such as:

- crime_hour
- day_type

- year
- month

• Categorizing crime descriptions into broader **crime categories** for easier analysis
• Creating **victim_age_category** groups for demographic analysis.

4. Exploratory Data Analysis

Several key analytical questions were explored:

1. What is the total number of reported crimes?
2. Which areas experience the highest crime activity?
3. What types of crimes occur most frequently?
4. Which weapons are most commonly used in crimes?
5. What is the demographic distribution of victims by age group?
6. How does crime activity change over time?
7. At what hours of the day do crimes occur most frequently?

5. Power BI Dashboard

An interactive **Crime Analytics Dashboard** was created using Power BI.

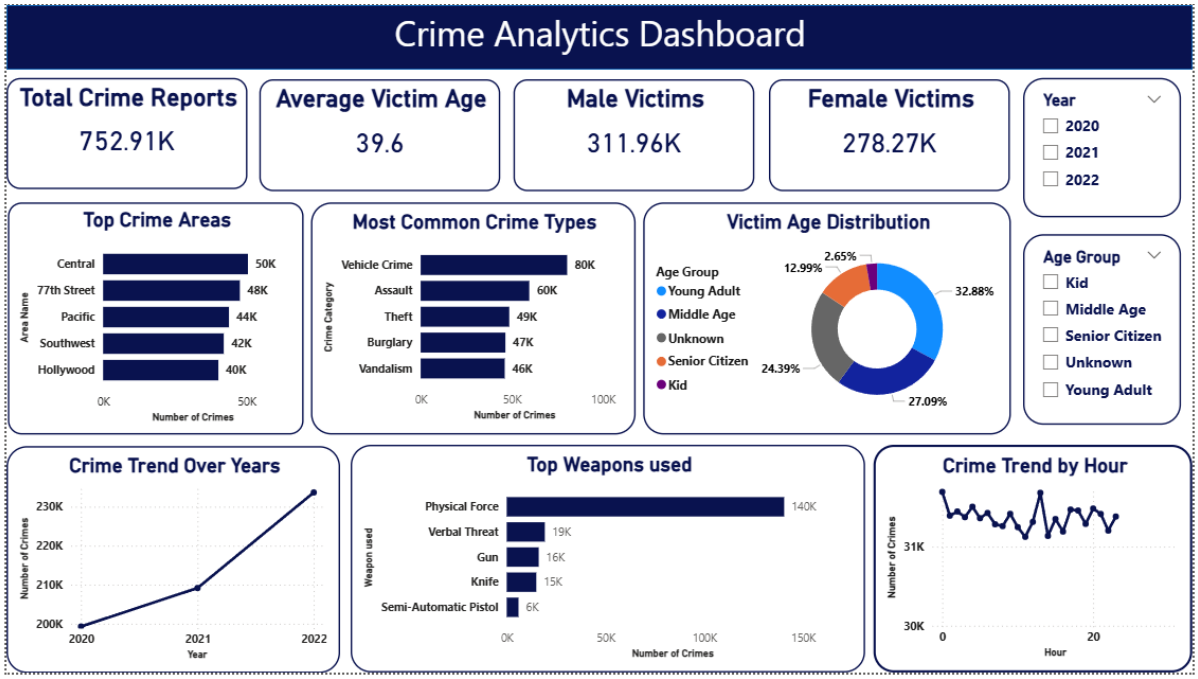
The dashboard includes the following visual insights:

• Total crime reports
• Average victim age
• Male vs female victim counts
• Top crime areas
• Most common crime categories
• Victim age distribution
• Top weapons used in crimes
• Crime trend over years
• Crime distribution by hour of day

Interactive slicers allow filtering by:

- Year
- Victim Age Group

This enables users to explore crime patterns dynamically.



6. Key Insights

Key findings from the analysis include:

- Certain areas report significantly higher crime volumes than others.
- Vehicle-related crimes and assaults are among the most common offenses.
- Most victims fall within the **young adult and middle-age groups**.
- Crimes involving **physical force** appear more frequently than firearm-related incidents.